EXPERIMENT - 6

AIM: To create a stored procedure transfer_employee(emp_id, new_dept_id) with validation and error handling, and to implement triggers for salary validation and audit logging, including testing of edge cases.

SOURCE CODE:

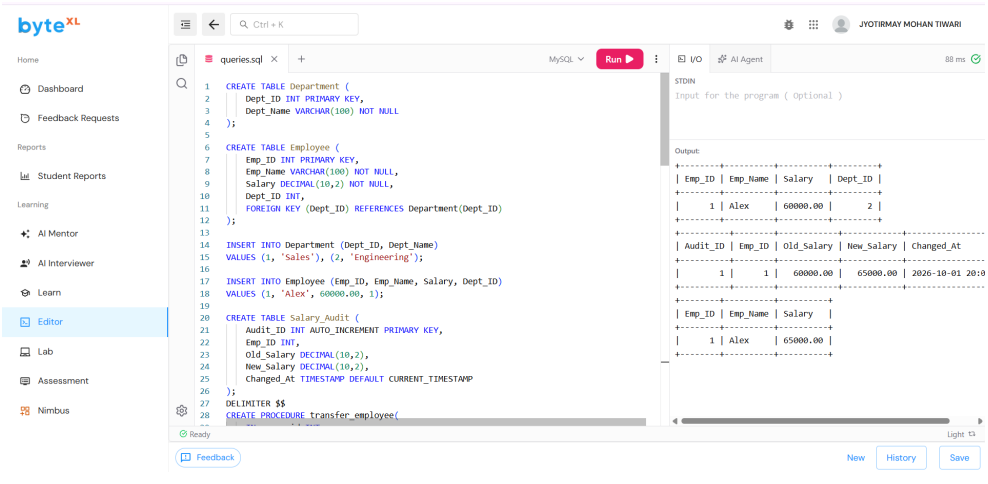
```
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(100) NOT NULL
);
CREATE TABLE Employee (
  Emp_ID INT PRIMARY KEY,
  Emp_Name VARCHAR(100) NOT NULL,
  Salary DECIMAL(10,2) NOT NULL,
  Dept_ID INT,
  FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID)
);
INSERT INTO Department (Dept_ID, Dept_Name)
VALUES (1, 'Sales'), (2, 'Engineering');
INSERT INTO Employee (Emp_ID, Emp_Name, Salary, Dept_ID)
VALUES (1, 'Alex', 60000.00, 1);
CREATE TABLE Salary_Audit (
  Audit_ID INT AUTO_INCREMENT PRIMARY KEY,
  Emp_ID INT,
  Old_Salary DECIMAL(10,2),
  New_Salary DECIMAL(10,2),
  Changed_At TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
DELIMITER $$
CREATE PROCEDURE transfer_employee(
```

```
    IN p_emp_id INT,
    IN p_new_dept_id INT
)
BEGIN
    DECLARE emp_count INT DEFAULT 0;
    DECLARE dept_count INT DEFAULT 0;
    SELECT COUNT(*)
    INTO emp_count
    FROM Employee
    WHERE Emp_ID = p_emp_id;
    SELECT COUNT(*)
    INTO dept_count
    FROM Department
    WHERE Dept_ID = p_new_dept_id;
    IF emp_count = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Employee does not exist';
    ELSEIF dept_count = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Department does not exist';
    ELSE
        UPDATE Employee
        SET Dept_ID = p_new_dept_id
        WHERE Emp_ID = p_emp_id;
    END IF;
END $$
DELIMITER ;
CALL transfer_employee(1, 2);
SELECT *
```

```
FROM Employee
WHERE Emp_ID = 1;
-- These calls intentionally raise errors; uncomment and test them individually.
-- CALL transfer_employee(999, 2);
-- CALL transfer_employee(1, 999);
DELIMITER $$
CREATE TRIGGER before_employee_salary_update
BEFORE UPDATE ON Employee
FOR EACH ROW
BEGIN
  IF NEW.Salary < 0 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Salary cannot be negative';
  END IF;
END $$
DELIMITER ;
-- This update intentionally raises an error; uncomment to test the validation trigger.
-- UPDATE Employee SET Salary = -5000 WHERE Emp_ID = 1;
DELIMITER $$
CREATE TRIGGER after_employee_salary_update
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
  IF OLD.Salary <> NEW.Salary THEN
    INSERT INTO Salary_Audit
    (
      Emp_ID,
      Old_Salary,
      New_Salary
```

```
    )
    VALUES
    (
      OLD.Emp_ID,
      OLD.Salary,
      NEW.Salary
    );
  END IF;
END $$
DELIMITER ;
UPDATE Employee
SET Salary = 65000
WHERE Emp_ID = 1;
SELECT *
FROM Salary_Audit;
SELECT
  Emp_ID,
  Emp_Name,
  Salary
FROM Employee
WHERE Emp_ID = 1;
```

OUTPUT –




byteXL
Home
Dashboard
Feedback Requests
Reports
Student Reports
Learning
AI Mentor
AI Interviewer
Learn
Editor
Lab
Assessment
Nimbus
Ctrl + K
JYOTIRMAY MOHAN TIWARI
I/O
AI Agent
88 ms
STDIN
Input for the program (Optional)
Output:
+--------+----------+----------+---------+
| Emp_ID | Emp_Name | Salary | Dept_ID |
+--------+----------+----------+---------+
| 1 | Alex | 60000.00 | 2 |
+--------+----------+----------+---------+
+----------+--------+------------+------------+---------------------+
| Audit_ID | Emp_ID | Old_Salary | New_Salary | Changed_At |
+----------+--------+------------+------------+---------------------+
| 1 | 1 | 60000.00 | 65000.00 | 2026-10-01 20:02:58 |
+----------+--------+------------+------------+---------------------+
+--------+----------+----------+
| Emp_ID | Emp_Name | Salary |
+--------+----------+----------+
| 1 | Alex | 65000.00 |
+--------+----------+----------+
Ready
Light
Feedback
New
History
Save